NPTEL ONLINE CERTIFICATION COURSES

Course Name: Software Project Management

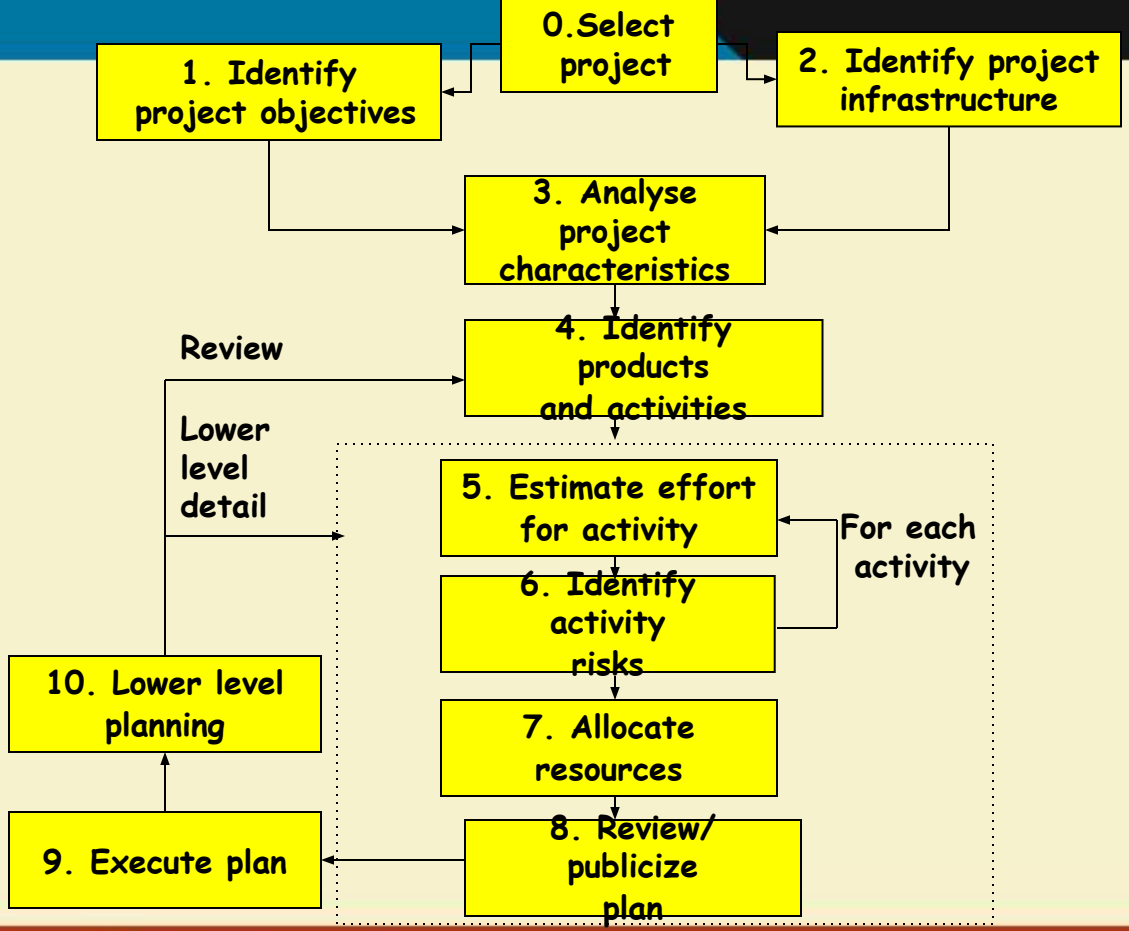
Faculty Name: RAJIB MALL

Department : Computer Science and Engineering

TOPIC: Project Scheduling

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Project Steps



Project Scheduling

Project Scheduling Process

1. Clearly define project objective (scope)
2. Develop a Work Breakdown Structure (WBS)
3. Identify precedence relationships
4. Estimate time needed for each task
5. Determine resources needed for each task
6. Draw PERT diagram

Project Scheduling Process

7. Calculate *earliest* and *latest starting times*, *earliest* and *latest completion times, slack* times, *critical path* etc.

8. Construct a GANTT chart.

9. Reallocate resources if necessary.

10. Continuously monitor/revise the time estimates along the project duration.

What is an Activity?

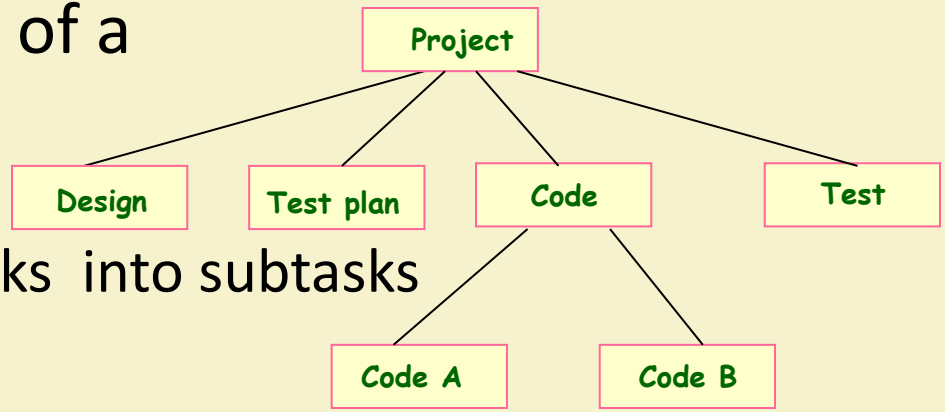
- **An activity:**
  - Must have a duration-- defined by start and end-points
  - Must have resource requirements: usually constant
  - May be dependent on other activities (precedence)

Identifying activities

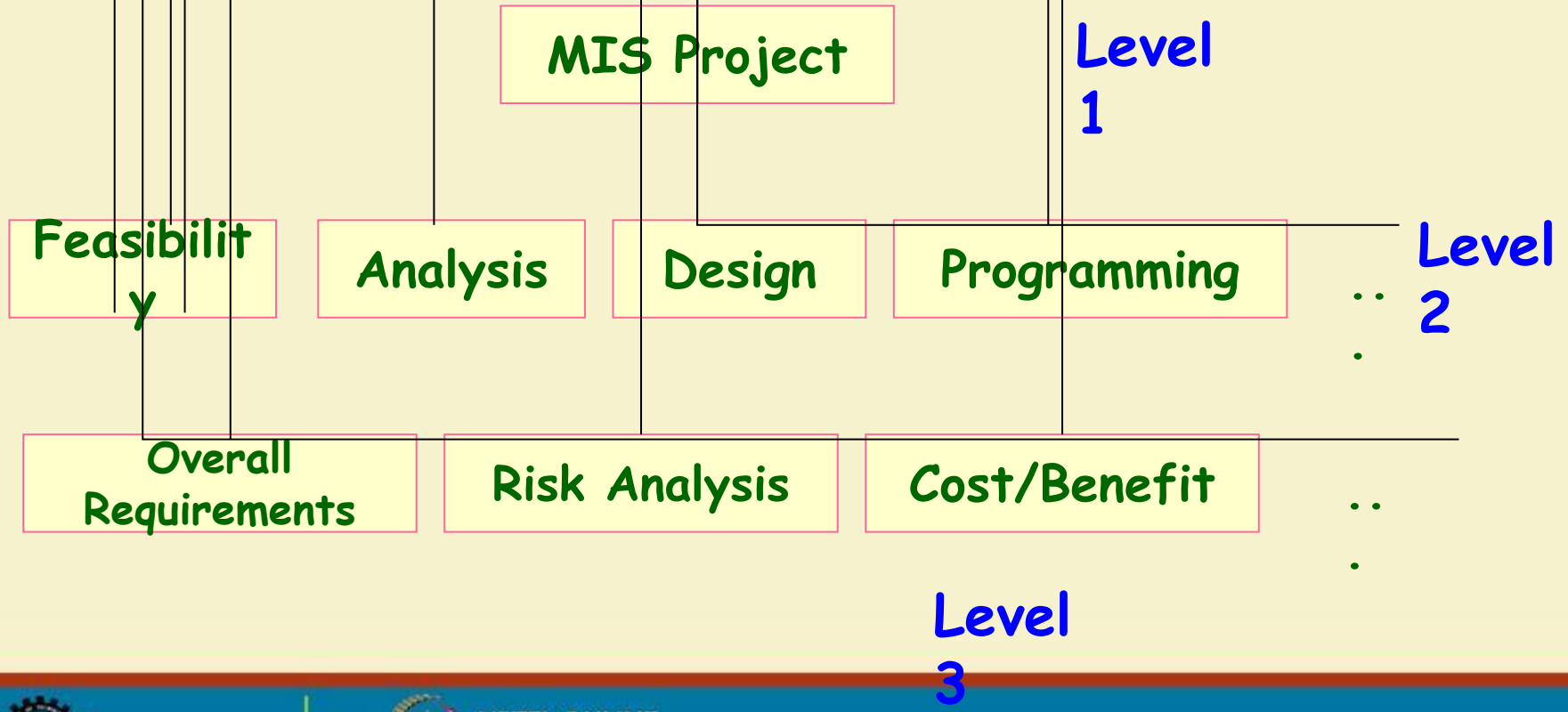
- **Work-based:**
  - Draw-up a Work Breakdown Structure listing the work items needed
- **Deliverable-based approach:**
  - List the deliverables of project
  - Identify the order in which these have to be created
  - Work out activities needed to create these

Work breakdown structure (WBS)

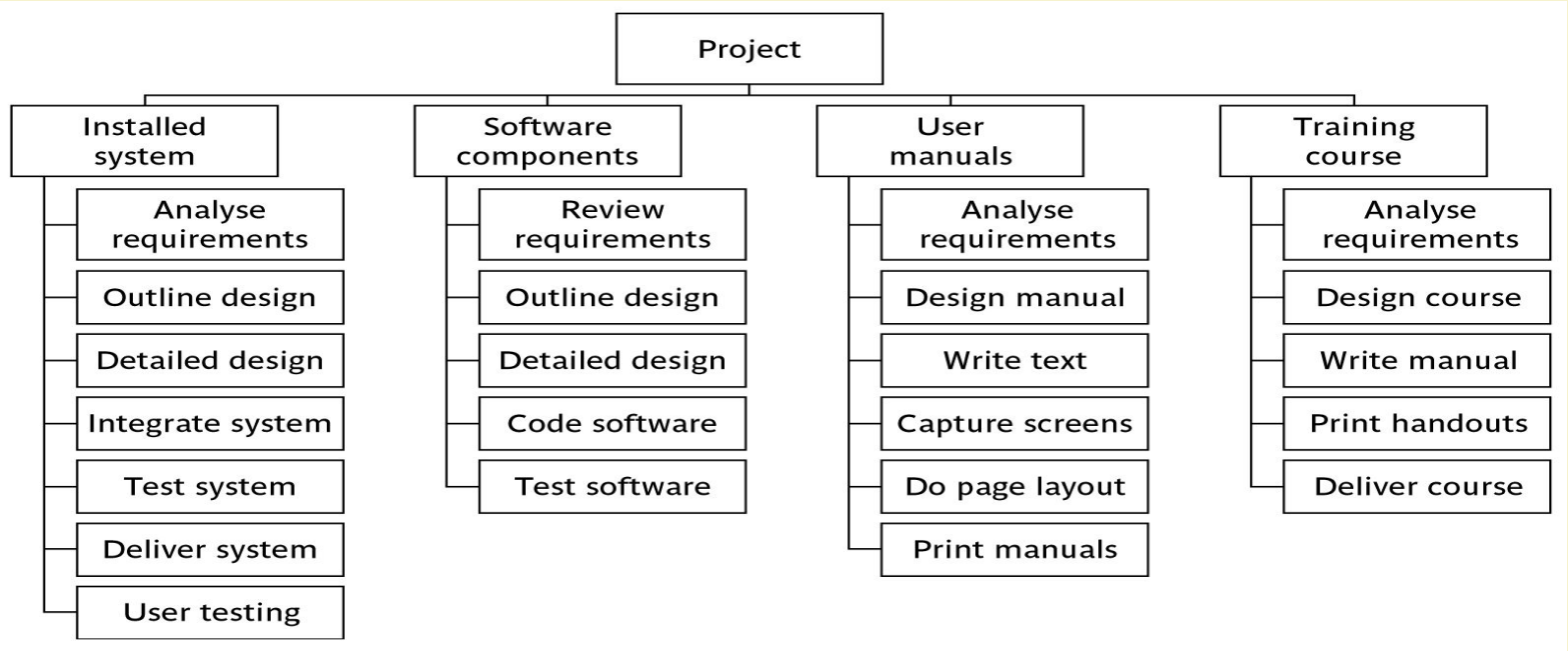
- Hierarchical decomposition of a project tasks into subtasks:
  - Shows decomposition of tasks into subtasks
  - Does not show duration
  - Does not show precedence relations (e.g. coding must be finished before Testing can start)



WBS Example

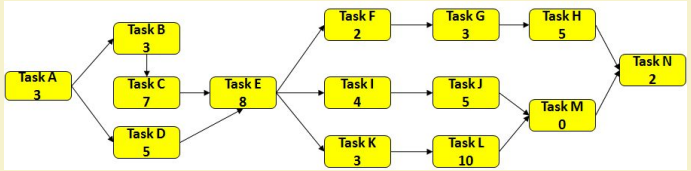

MIS Project
Level 1
Feasibility
Analysis
Design
Programming
Level 2
Overall Requirements
Risk Analysis
Cost/Benefit
Level 3

Hybrid approach



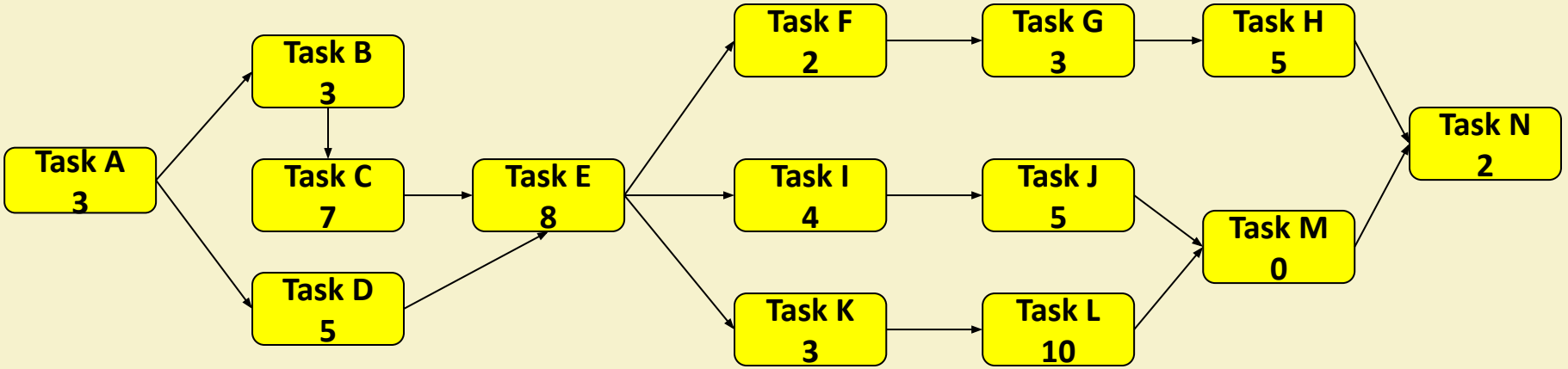
A Simple Task Network

- Also called an **Activity Network**
- Depicts task flow for a project
  - Task length, sequence, concurrency, and dependency
  - Represents inter-task dependencies



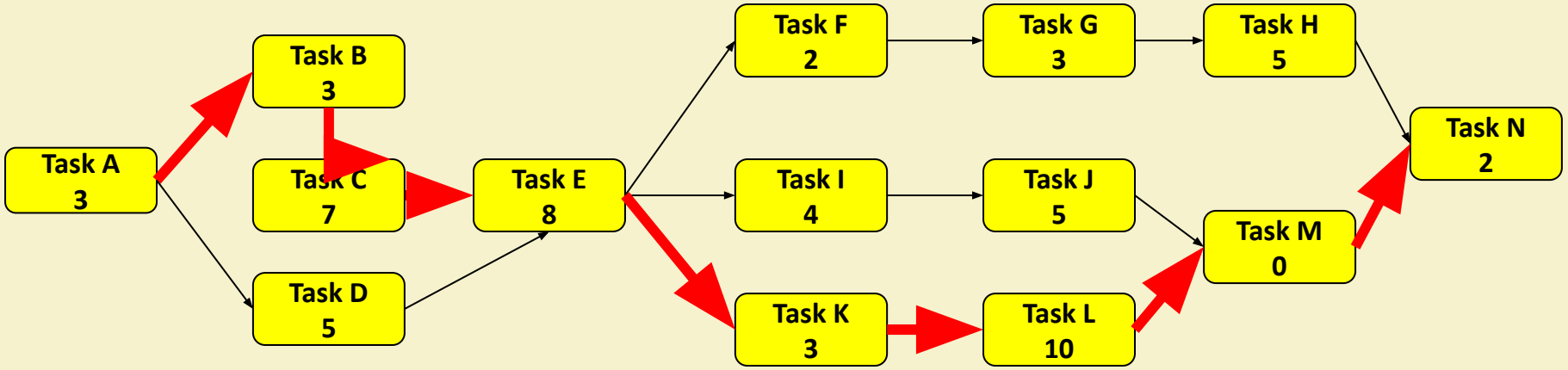
- **The critical path:**
  - **A single path leading from start to finish in a task network**
  - **Contains the sequence of tasks that must be completed on schedule, if the project as a whole is to be completed on schedule**
  - **Also determines the minimum duration of the project**

Example Task Network



- Which is the critical path and what tasks are on it?
- Is it possible to have two critical paths?

Example Task Network with Critical Path Marked



Critical path: A-B-C-E-K-L-M-N

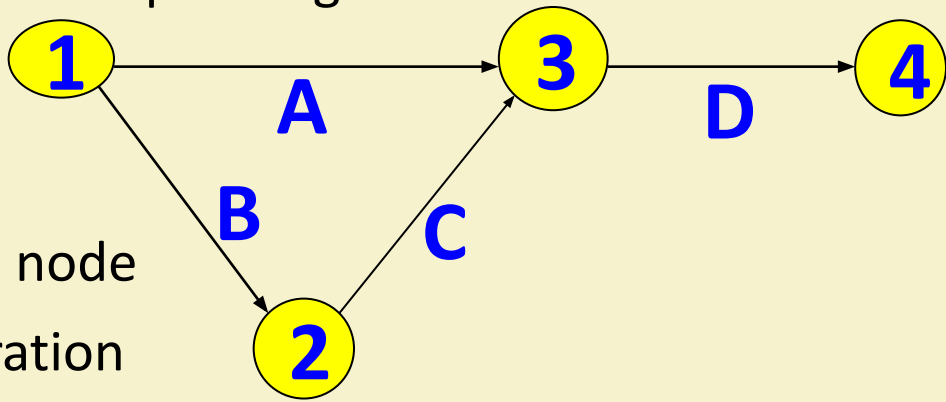
CONCEPTS COVERED

- Project Scheduling using PERT/CPM



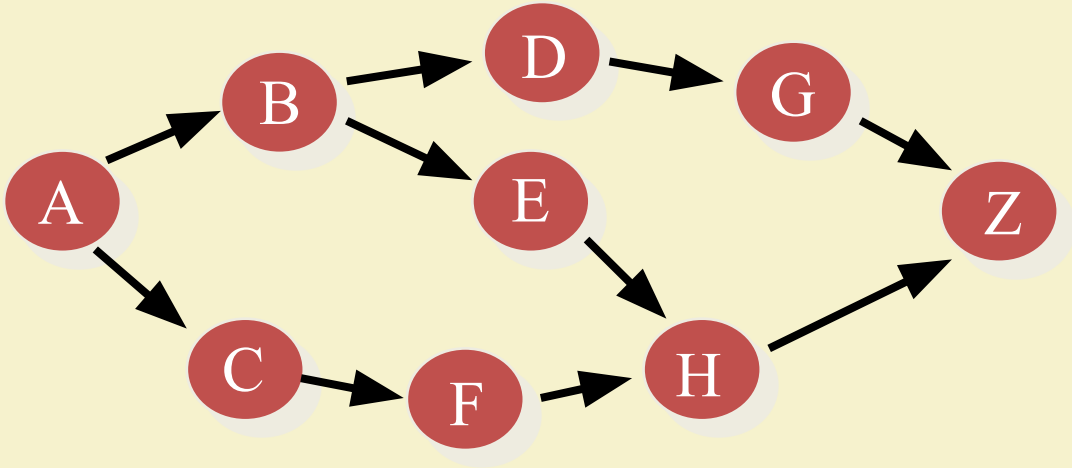
Task Networks

- **Activity-on-node diagrams:**
  - More than single start and end nodes may exist
  - However, recommended to use one start and one end node for clarity
  - Nodes represent activities and the corresponding durations
  - Arrows indicate precedence
- **Activity-on-arrow diagrams:**
  - One single start and one single end node
  - Arrows represent activities and duration
  - Nodes indicate beginning/end of activities



Activity on Node
Task Network Example

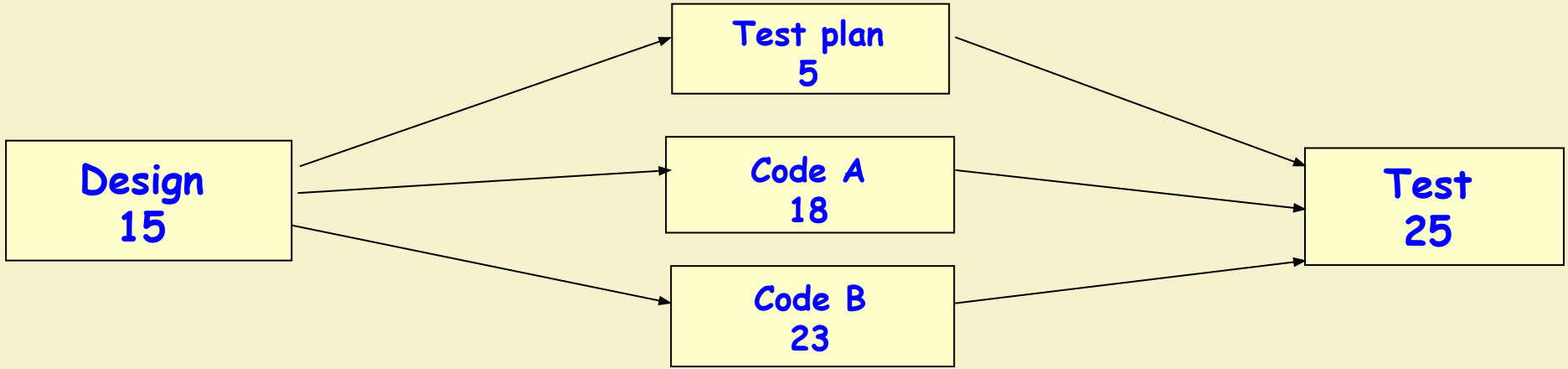
Activity	Predecessors
A	--
B	A
C	A
D	B
E	B
F	C
G	D
H	E, F



Two Task Networks: PERT & CPM

- **Program Evaluation and Review Technique and Critical Path Method:** Task Network-based techniques
- Developed in 1950's
  - **CPM by DuPont for chemical plants**
  - **PERT by U.S. Navy for Polaris missile**
- Considers precedence relationships & interdependencies
- Each uses a different estimate of activity times

PERT (Program Evaluation and Review Technique)



- A network in which nodes represent tasks and arrows describe precedence relations
  - Used successfully in management of Polaris missile project in 1950's
  - Shows task duration (on the task node)

PERT Chart

- Multiple time estimates were used for each task:
  - Allowed for variations due to uncertainty in determined activity times
  - Activity times are assumed to be random, with certain assumed probability distribution
- Originally was an *activity on arc* network:
  - Over time, people began to use PERT as an *activity on node* network.

Probabilistic Time Estimates

- PERT uses 3 time estimates for each activity:
- **Most likely time (m)**
  - subjective estimate of most frequent time
- **Optimistic time (a)**
  - shortest possible time (ideally)
- **Pessimistic time (b)**
  - longest time possible if everything went wrong

CPM

- Developed to manage maintenance projects in the chemical industry:
  - **A complex undertaking, but individual tasks are routine ----> task duration is deterministic**
  - **Did not support variation in activity times**
  - **Activities are represented as rectangles or circles**
  - **Allowed controlling project duration through identification of critical path**

- Graphically display the precedence relationships & sequence of activities

Both PERT and CPM

- Estimate the project's duration
- Identify **critical activities**:
  - Activities that cannot be delayed without delaying the project
- Estimate the amount of slack associated with non-critical activities

- Over time, CPM and PERT merged into one technique:
  - referred to as "**CPM/PERT**".

CPM/PERT

- It is easy to see task precedence relationships
- Ideal for large projects with many activities
  - Consist of a network of branches and nodes

Sample Usage of PERT & CPM

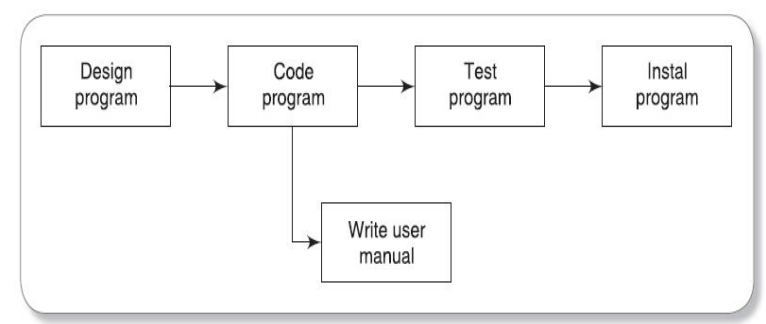
- Completion date?
- On schedule? Within budget?
- Probability of completing by ...?
- Critical activities?
- Impact of resources availability?

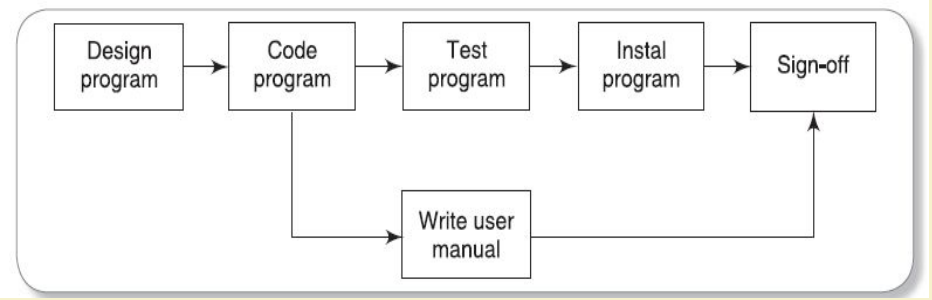
PERT & CPM Steps

- Identify activities
- Determine sequence
- Create network
- Determine activity times
- To find critical path:
  - **Determine Earliest & latest start times**
  - **Determine Earliest & latest finish times**
  - **Determine Slack**

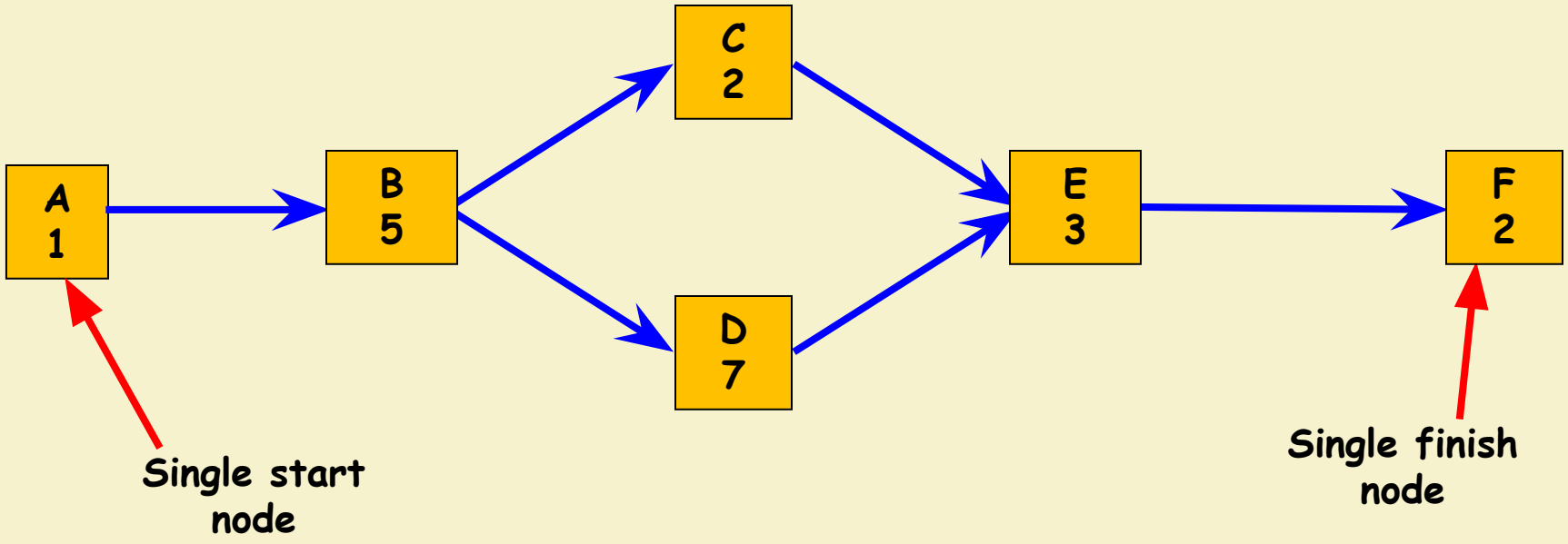
Task Network Guidelines

- A project can have only one start and one end node
- A network may not contain loops
- A network should not contain dangles

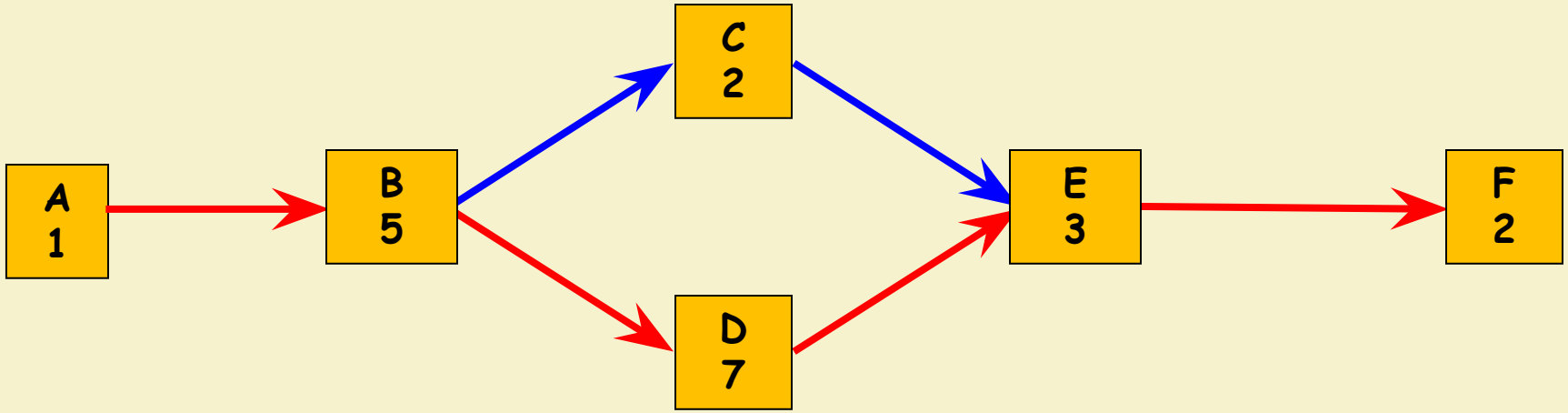




Example PERT Diagram



Critical Path



Activity Characterization

- **Earliest Start (ES)** = Earliest finish of the immediately preceding activity
- **Earliest Finish (EF)** = **ES plus** the **activity time**
- **Latest Start (LS) and Latest Finish (LF)** = The latest an activity can start (LS) or finish (LF) without delaying the project completion
- **Slack defines how long non-critical activities can be delayed without delaying the project**
- **Slack** = Activity's **late finish minus its early finish** (or its **late start minus its early start**)

Example: Slack or Float

- Earliest start = day 5
- Latest finish = day 30
- Duration = 10 days

- **Earliest finish =?**
- **Latest start = ?**

Float = LF - ES - duration

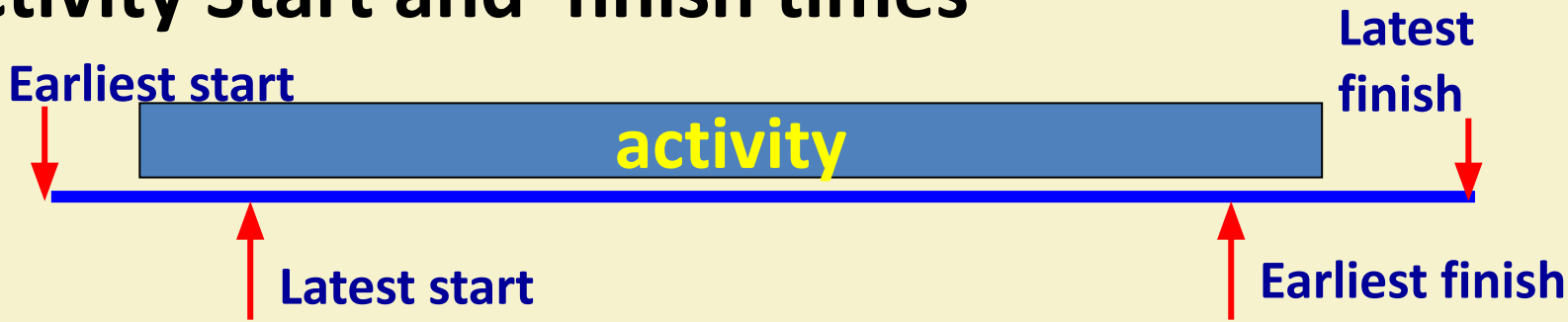
What is the float in this case?

CONCEPTS COVERED

- Computation of Project Characteristics using PERT/CPM

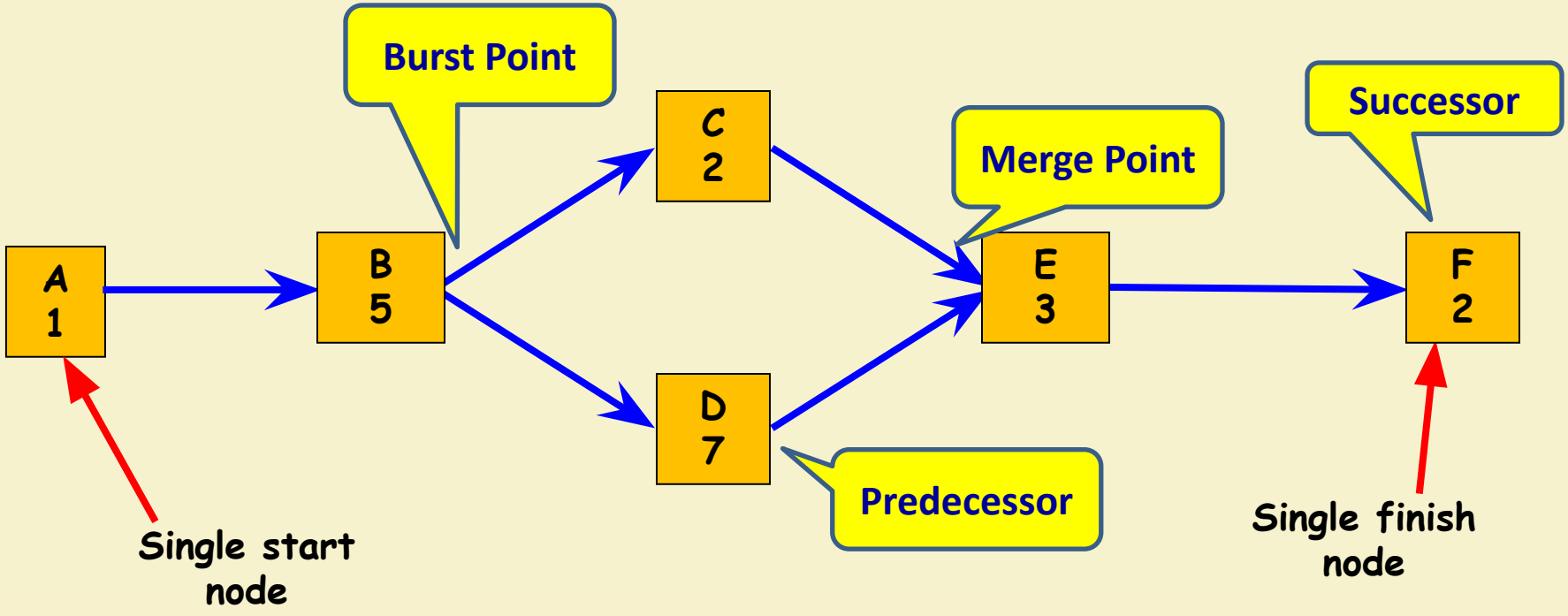


Activity Start and finish times

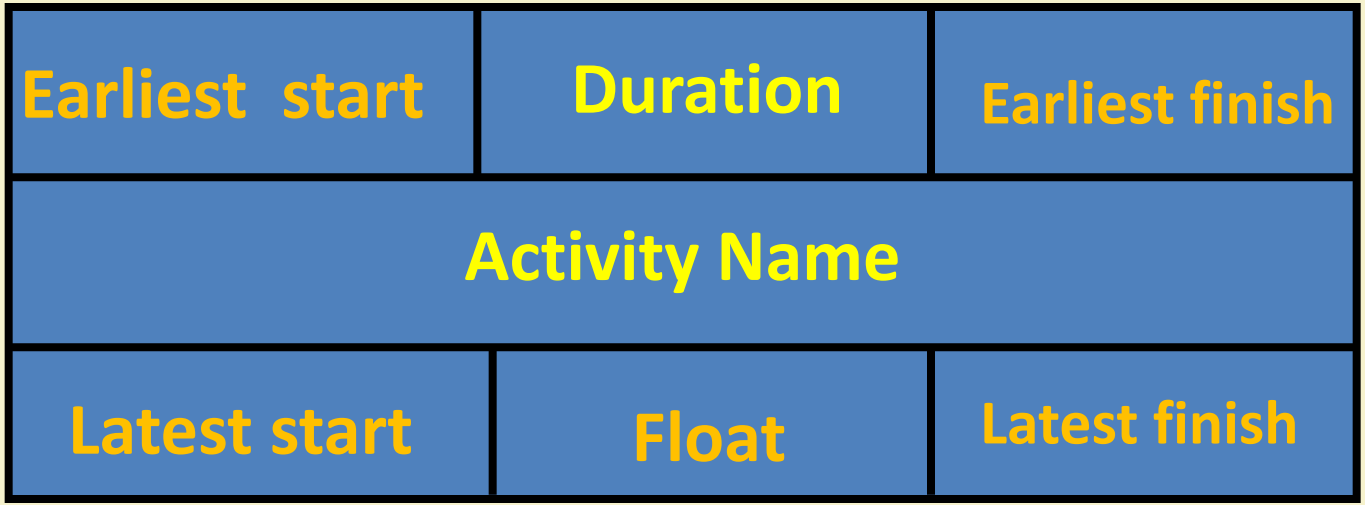


- Activity ‘write software documentation’ = Duration 7 days
- Earliest start (ES)
- Earliest finish (EF) = ES + duration
- Latest finish (LF) = latest task can be completed without affecting project completion time
- Latest start = LF - duration

PERT/CPM Terminologies



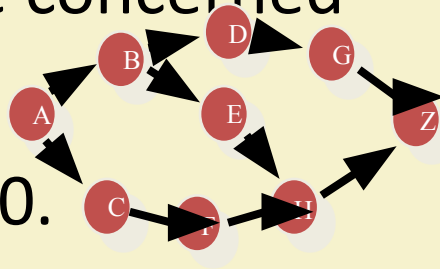
Task Characteristics Computed Using PERT/CPM



- Day numbers used rather than actual dates

Day 0

- Makes initial calculations easier – need not be concerned with week-ends and public holidays



- **Finish** date Day 10 means at the END of Day 10.
- **Start** date Day 1 also means at the END of Day 1.
- First activity conventionally begins at Day 0 i.e. end of Day 0 or start of Day 1

Importance of Float (Slack) and Critical Path

- Slack or Float shows how much flexibility each activity has:
  - **That is, how long it can be delayed without affecting completion date**
- Critical path is a sequence of activities from start to finish with zero slack.
  - Critical activities are activities on the critical path.
- **Critical path identifies the minimum time to complete project**
  - If any activity on the critical path is delayed project will be delayed

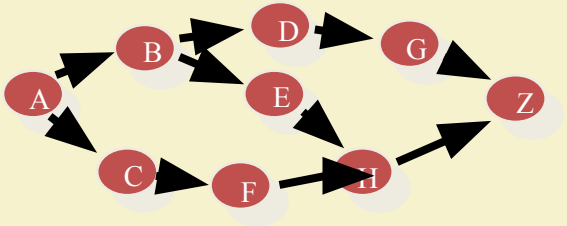
Computing project parameters using PERT/CPM chart

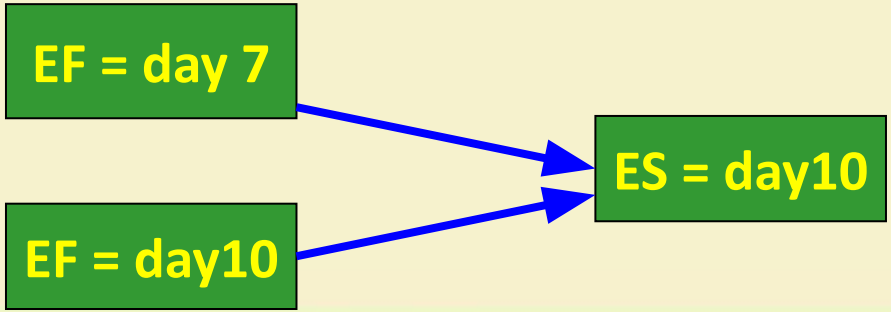
- Forward pass
- Backward pass
- Identify critical paths



Forward pass

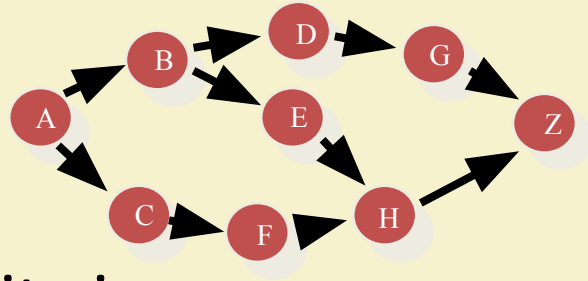
- Start at beginning (Day 0) and work forward following chains.
- Earliest start date for the *current* activity = earliest finish date for the *previous*
- When there is more than one previous activity, take the *latest* earliest finish





Backward Pass

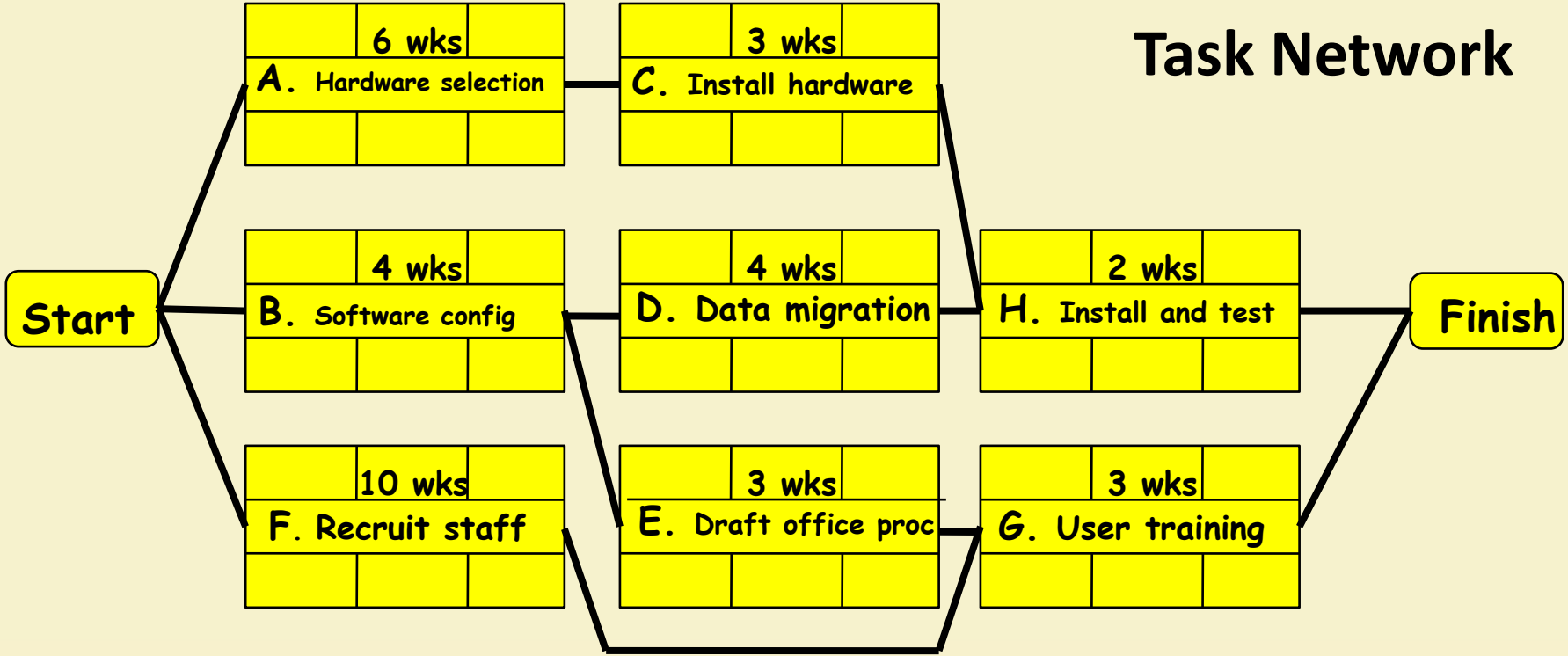
- Latest completion time (LC) of an activity:
  - Latest completion time for any activity is equal to latest start time of its successor.
  - If more than one successors, smallest of the latest start times of the activity's immediate successors.



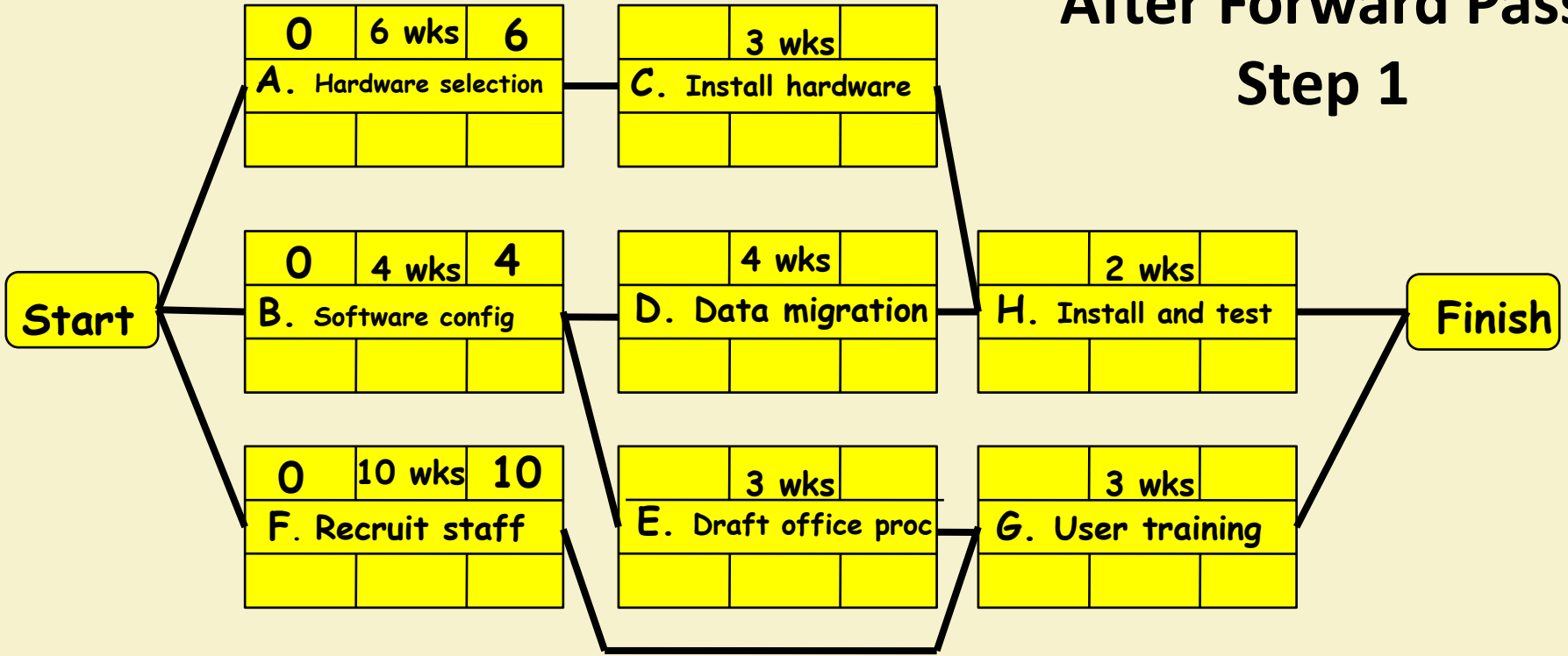
Example

Activity Label	Activity Name	Duration (Weeks)	Precedence
A	Hardware Selection	6	---
B	System configuration	4	---
C	Install hardware	3	A
D	Data migration	4	B
E	Draft office procedures	3	B
F	Recruit staff	10	---
G	User training	3	E,F
H	Install and test	2	C,D

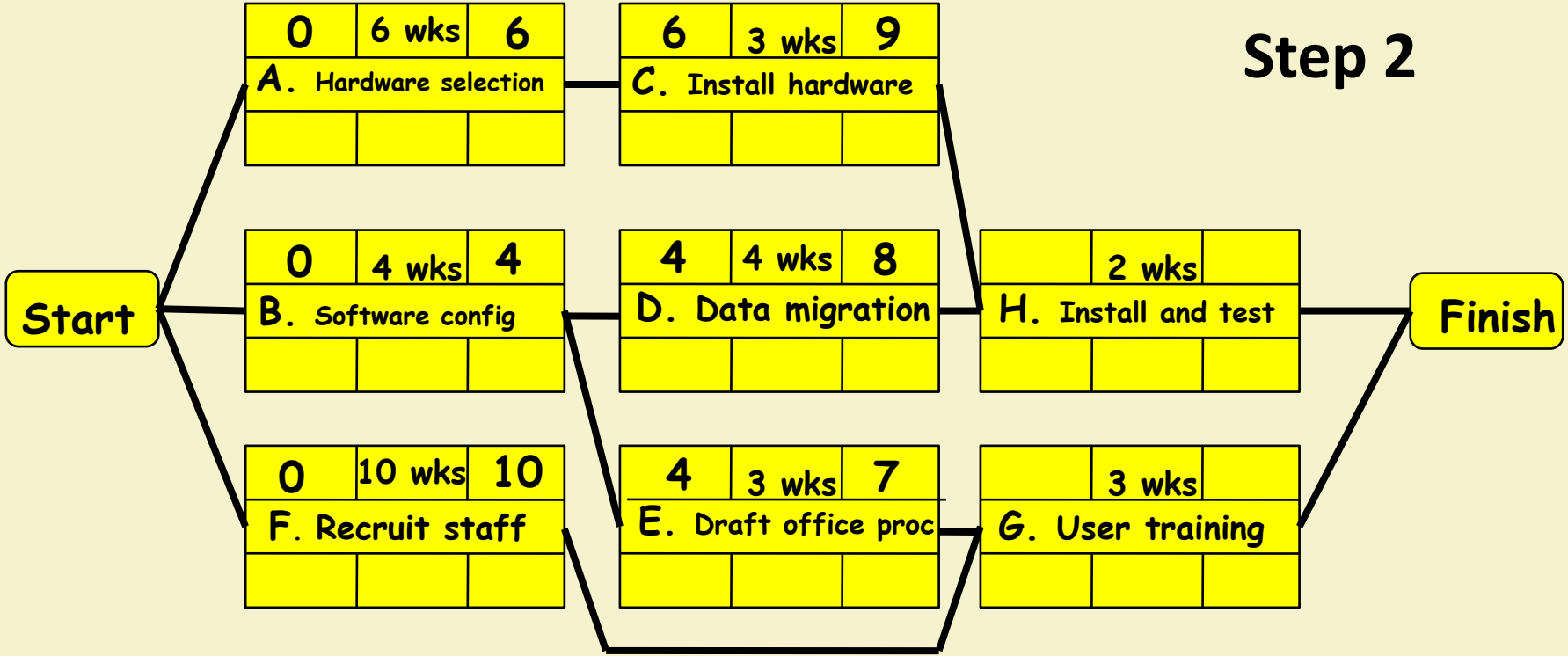
First Draw Task Network



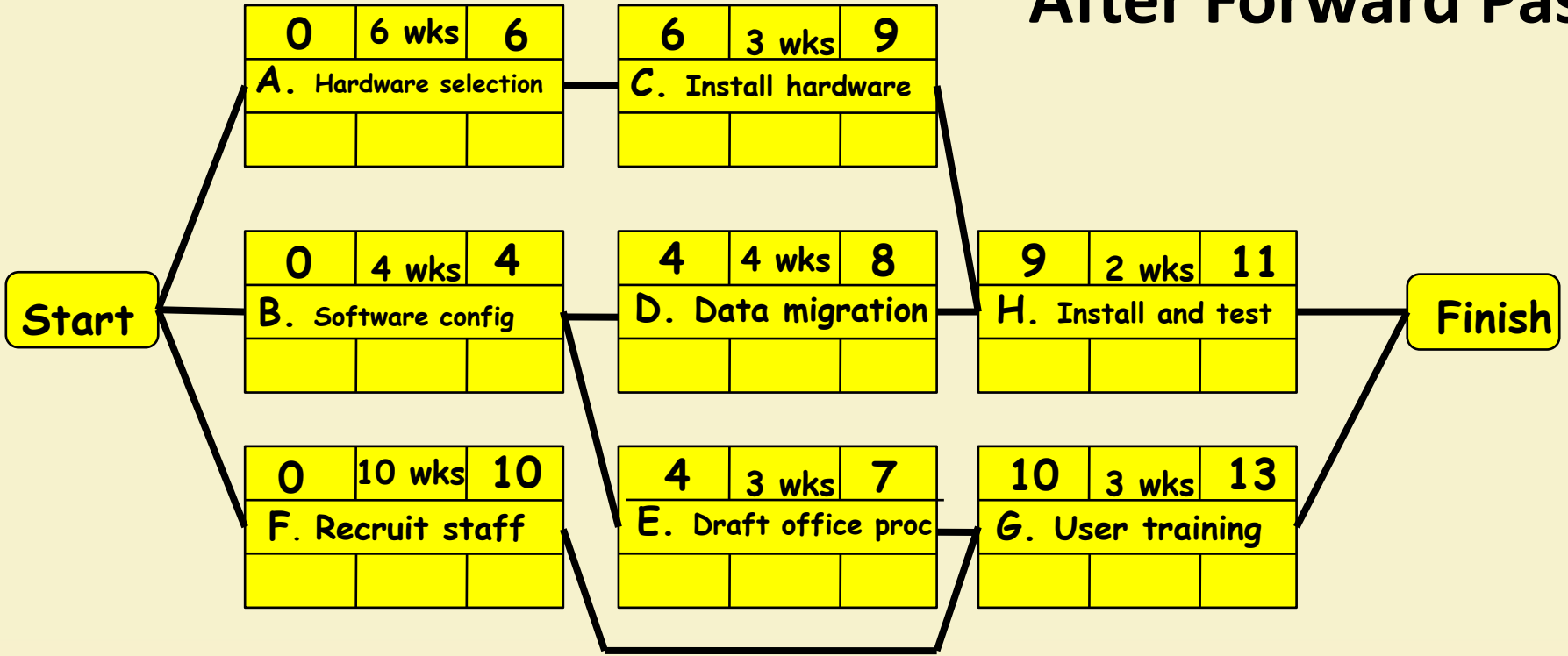
After Forward Pass Step 1



After Forward Pass Step 2



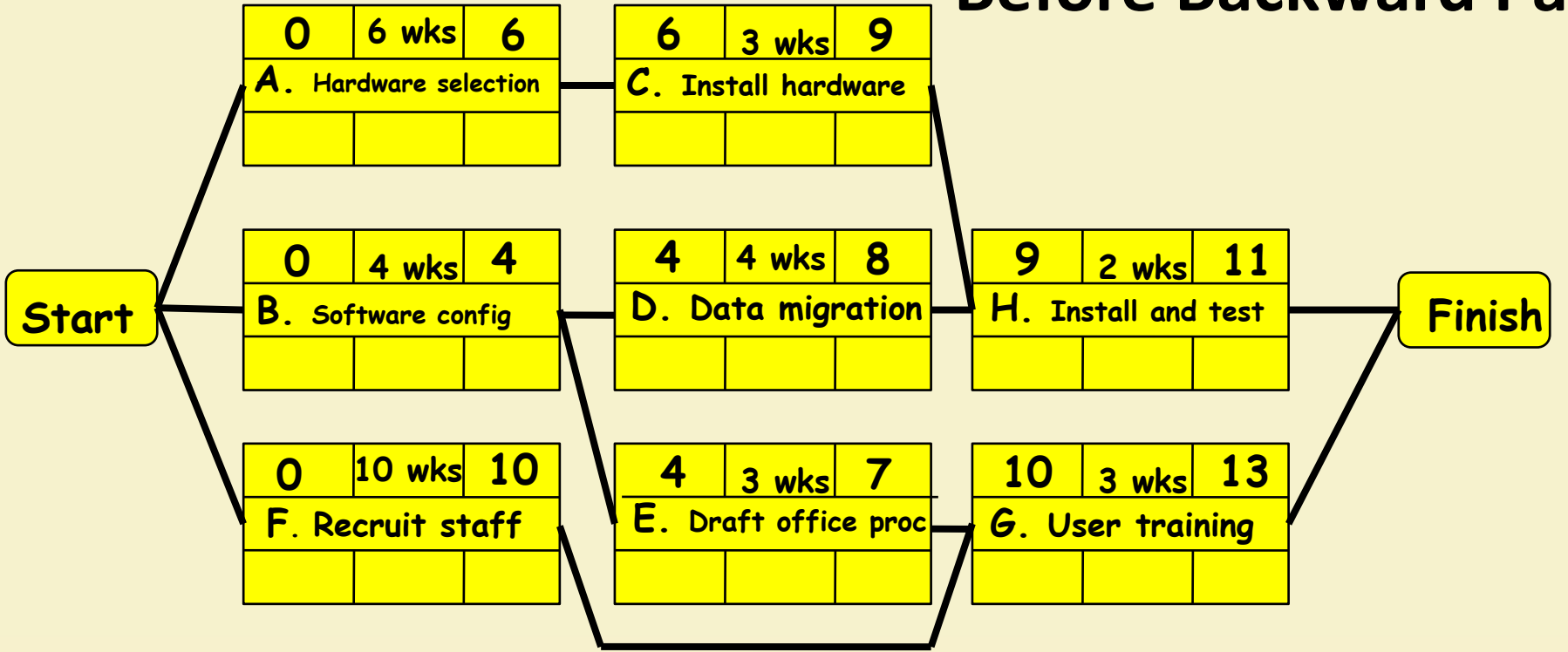
After Forward Pass



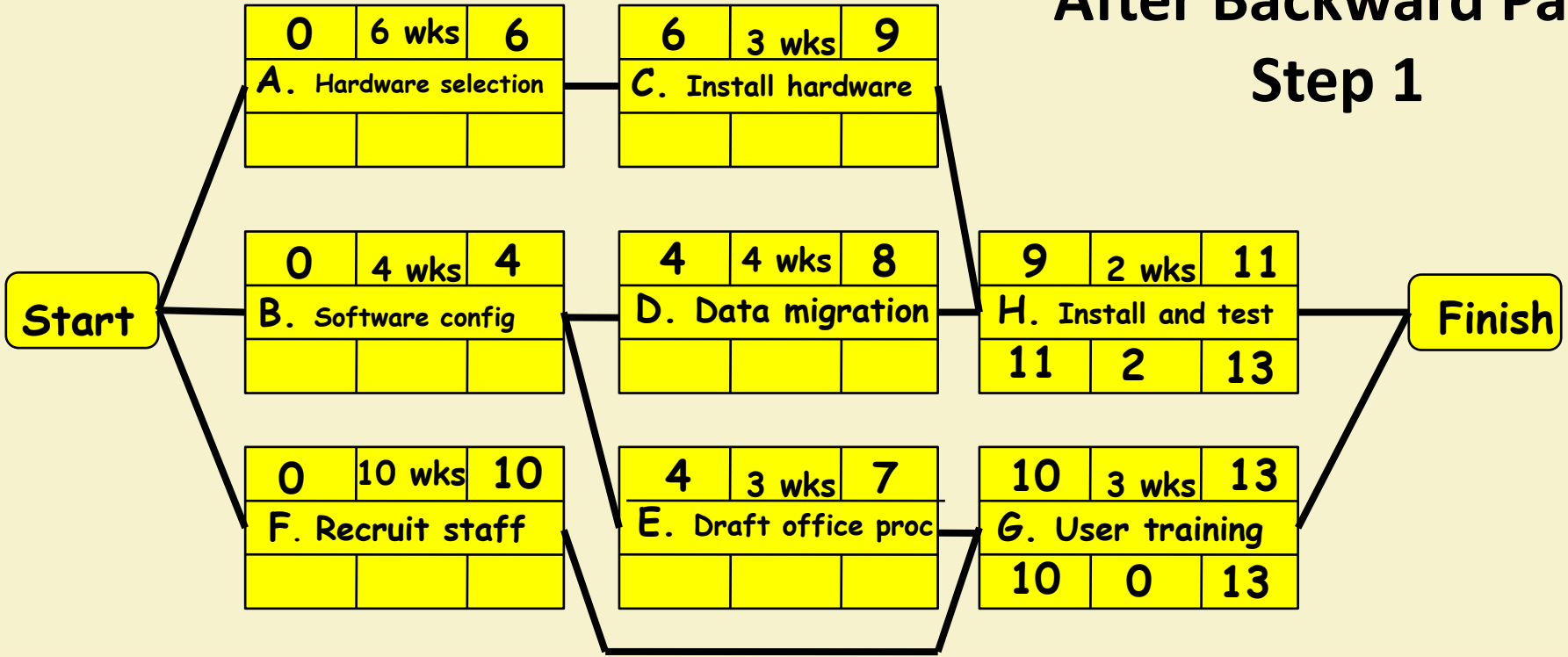
Backward pass

- Start from the *last* activity
- For last activity Latest finish (LF) = earliest finish (EF)
- Work backwards
- Latest finish for *current* activity = Latest start for the *following*
- More than one following activity □ take the *earliest* LS
- Latest start (LS) = LF for activity - duration

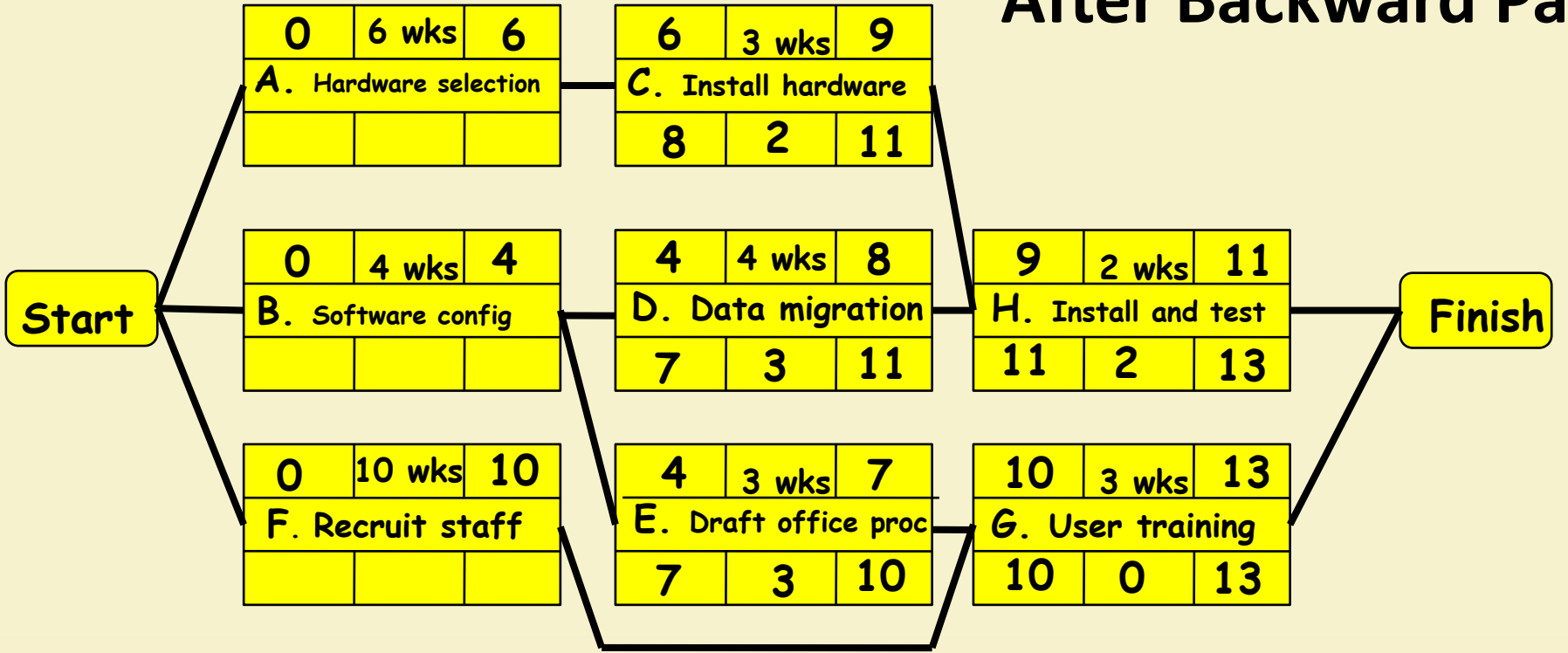
Before Backward Pass



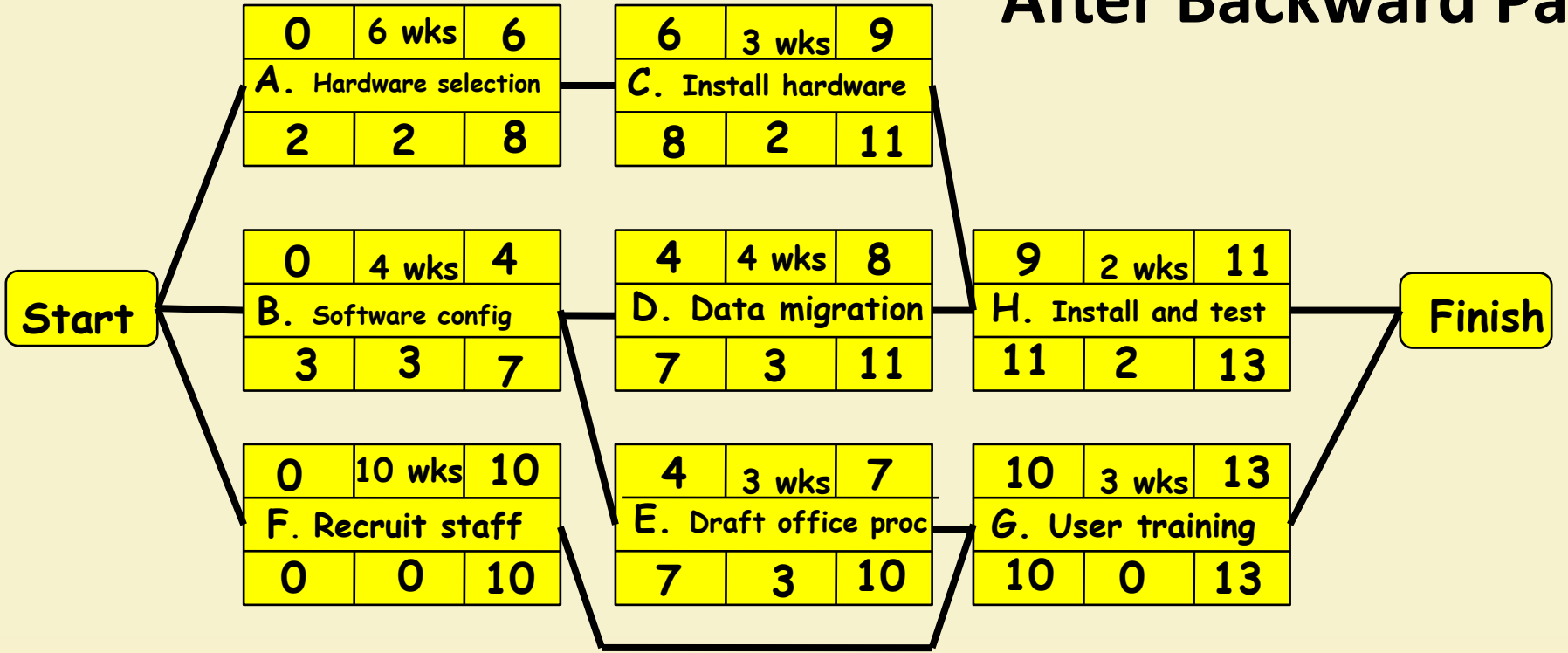
After Backward Pass Step 1



After Backward Pass



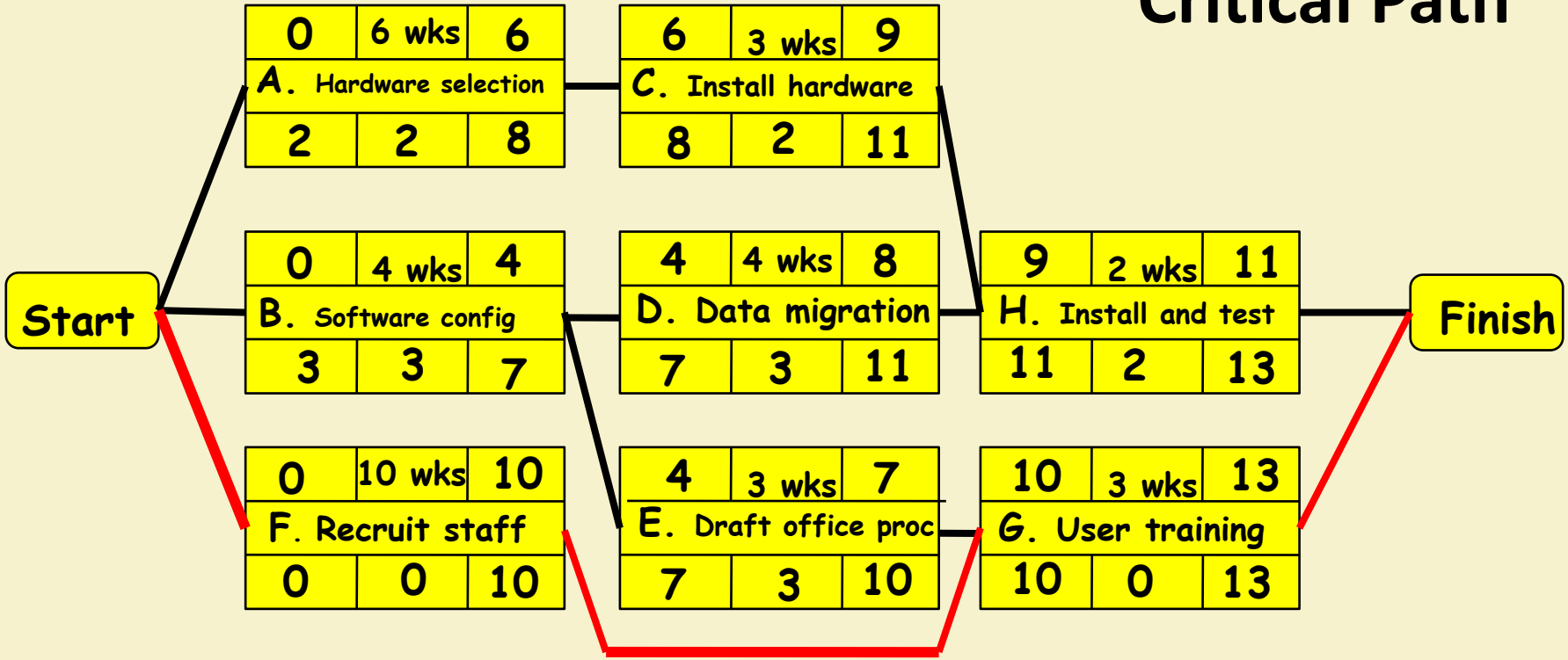
After Backward Pass



Activity	ES	Dur	EF	LS	LF	Float
A						
B						
C						
D						
E						
F						
G						

Complete the table

Critical Path



Critical path

- It is the path through network with zero floats
- Any delay in an activity on the critical path will delay whole project
- Can there be more than one critical path?
- Can there be no critical path?
- Sub-critical paths

Scheduling Exercise 1

Activity	Immediate predecessors	Completion Time (week)
A	-	5
B	-	6
C	A	4
D	A	3
E	A	1
F	E	4
G	D,F	14
H	B,C	12
I	G,H	2

Scheduling Exercise 2

Activity	Predecessor	Duration
A	-	2
B	-	6
C	-	4
D	A	3
E	C	5
F	A	6
G	B, D, E	3


Thank you